Giulio Cusenza

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Google Scholar | GitHub | LinkedIn



Education

present **M.Sc. in AI & Machine Learning @ Technical University Darmstadt**
GPA: **1.33** (**A**). Studying AI with a focus on **language** and **multimodal** architectures.

2024 **B.A. in Computational Linguistics @ University of Tübingen**
Graduated with **1.18** (**A+**). Thesis: "*On the difficulty of using Contextual Word Embeddings to Measure Political Polarisation in Parliamentary Speech*" (link).

2021 **Classical high school degree @ Liceo Classico Umberto I**
Graduated with **100/100** (key subjects: Latin, Ancient Greek, Philosophy) in Palermo, Italy.

Experience

Oct 2026 – Mar 2027 **Research Assistant @ Technical University Darmstadt, Multimodal AI Lab**
Working on efficient **latent video representations** with Mohamed Abdelsalam and Marcel Klemt.

Jan 2024 – May 2024 **Intern in Data & Technology @ MHP – A Porsche Company**
Automated company processes by applying **Machine Learning** and **NLP** to **Robotic Process Automation** with UiPath.

Oct 2022 – Dec 2023 **Programming teaching assistant @ University of Tübingen**
Tutored over 70 students in the courses "*Data structures and algorithms for computational linguistics*" **I**, **II** (**Java**) and **III** (**python**).

Apr 2023 – Sep 2023 **Research assistant @ University of Tübingen, *"WORDS, BONES, GENES, TOOLS"* Center**
Worked on the **automatic identification** of **language borrowing** and **contact** through NLP and bioinformatic methods.

Skills

IT

- **Python** (**torch, transformers**, **spacy**, **nltk**).
- Experience with web: **HTML**, **CSS**, **JS**, Firebase (example).
- **Git** & **GitHub**.
- Familiar with **Java**, **Rust**, **bash**, **R**.
- Media: **Microsoft Office**, **Google Workspace**, **LaTeX**.

Languages

- Spoken: **Italian** (native), **English** (C2), **French** (C1), **Russian** (B2+), **German** (B2+).
- Heritage: **Sicilian**, **Arbëresh**.
- Written: **Latin**, **Ancient Greek**.
- Basic: Spanish, Korean.

Traits: curious | creative | detail-focused | committed | communicative

Publications

- Giulio Cusenza and Çağrı Çöltekin. 2024. NLP for Arbëresh: How an Endangered Language Learns to Write in the 21st Century. In *Proceedings of the 3rd Annual Meeting of the Special Interest Group on Under-resourced Languages @ LREC-COLING 2024*, pages 252–256, Torino, Italia. ELRA and ICCL.

Projects

Arbor

*Website with self-developed **NLP tools** (spellchecking, conjugation…) for the extremely low-resourced **Arbëresh** language spoken in Southern Italy.*

link

Polarisation & Contextual Embeddings

*Bachelor thesis on the challenges of measuring **parliamentary polarisation** through contextual word embeddings.*

GitHub
link

Question tagger model

*Training of a **question labelling** model based on Hugging Face Transformers.*

link

FST spell checker

*Probabilistic spell checker built on **weighted finite state transducers** (python implementation).*

GitHub
link

Awards

Deutschlandstipendium 2023 and 2026
(public scholarship)

"The Deutschlandstipendium supports talented and high-performing students. In addition to first-class grades, social commitment and personal achievements are also considered for the award."

Bundesministerium für Forschung, Technologie und Raumfahrt

link